

Annotated Bibliography

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[7] The article provides a summary of techniques presently used in sustainable development within the scientific community. The study begins by discussing the importance of sustainable development, the frequency of various techniques used to analyze sustainable development, and important issues in sustainable transportation, such as logistics and environmental issues. It then dives into the techniques that are currently being used to analyze sustainable development, which includes simulation, optimization, machine learning, and fuzzy set analysis. The Optimization section covered the history of optimization used in transportation and the variety of applications for optimization, including in the difference between urban transportation systems and in global/regional transportation systems. Applications in simulation focus on the growing integration of simulation and optimization models and on usages in logistics, traffic congestion, and on the usage of demand-based transportation systems (ride sharing applications). The machine learning section primarily summarizes a paper that uses a random forest model to understand patterns in a bike sharing transportation system, as well as touches on a variety of techniques (random forest, logistic regression, neural networks, support vector machines, decision trees, heuristics, multi-agent systems, and multi-variate analysis) for a variety of analytical, traffic-related outcomes. The section on fuzzy sets takes note of the wide range of uses that recent projects have undertaken utilizing fuzzy sets. The paper notes that sustainability factors into many of the research projects, where businesses seek sustainability to increase innovation and enhance competitiveness. The paper concludes by suggesting that more of the investigated technologies be applied in tandem with one another to increase effectiveness. The section on machine learning will be the most relevant to the IS, but the other sections provide relevant background information about other advances in the field as well as ways that these tools are used in the business world.

[8] The authors utilized a gradient-boosting machine learning algorithm in order to study traffic speeds in developing cities. By reading in data regarding road properties, day and time, and land use in the city of Cauayan in the Phillipines, the model predicted vehicle speeds in specific times and locations—generally finding that traffic speeds decreased near large urban areas. Importantly, the study notes that most similar studies into "smart city" development occur in urban settings, while this study provides information to policymakers from developing areas. The study provides a pertinent example of a machine learning algorithm being implemented to analyze "smart city" data to analyze urban growth in developing cities and enhance urban safety.

[3] This chapter provides a brief overview of the various techniques that combine GIS

data and machine learning currently employed in the civil engineering field. While it lists the name of the technique and a very brief paragraph about its recent advances, its upsides, and its downsides, the paper does not explain what the technique is or when it would be most beneficial to use it. There were also typos and incomplete sentences present in the writing. Neural networks, random forest and gradient boosting, support vector machines, decision trees, bayesian networks, gaussian processes, and ensemble learning all receive paragraph-long summaries. The paper then describes applications of machine learning to GIS, primarily in the analysis of environmental hazards (flood prediction, groundwater modeling, erosion, landslide susceptibility assessment, and LULCC) but also in areas such as public health, maintaining biodiversity, or in analyzing infrastructure costs. The paper concludes with an assessment that the combined use of GIS data and machine learning methods have the potential to benefit a large variety of fields. Although the paper encompasses a breadth of information about machine learning and GIS, it is held back by its brevity and will be more useful to me as a list of keywords to research in-depth elsewhere.

References

- [1] Ismail Abdulrashid, Wen-Chyuan Chiang, Jiuh-Biing Sheu, and Shamkhal Mammadov. An interpretable machine learning framework for enhancing road transportation safety. *Transportation Research Part E: Logistics and Transportation Review*, 195:103969, 2025.
- [2] Kenneth Li-Minn Ang, Jasmine Kah Phooi Seng, Ericmoore Ngharamike, and Gerald K. Ijamaru. Emerging technologies for smart cities’ transportation: Geo-information, data analytics and machine learning approaches. *ISPRS International Journal of Geo-Information*, 11(2), 2022.
- [3] N. R. Asha Rani, Sasmita Bal, and M. Inayathulla. Gis applications and machine learning approaches in civil engineering. In N. Vinod Chandra Menon, Sreevalsa Kolathayar, Hugo Rodrigues, and K. S. Sreekesava, editors, *Recent Advances in Civil Engineering for Sustainable Communities*, pages 157–166, Singapore, 2024. Springer Nature Singapore.
- [4] Ali Bahadori-Jahromi, Shah Room, Chia Paknahad, Marwah Altekreeti, Zeeshan Tariq, and Hooman Tahayori. The role of artificial intelligence and machine learning in advancing civil engineering: A comprehensive review. *Applied Sciences*, 15(19), 2025.
- [5] Thanapong Champahom, Chinnakrit Banyong, Thananya Janhuaton, Chamroeun Se, Fareeda Watcharamaisakul, Vatanavongs Ratanavaraha, and Sajjakaj Jomnonkwao. Deep learning vs. gradient boosting: Optimizing transport energy forecasts in thailand through lstm and xgboost. *Energies*, 18(7), 2025.
- [6] Yosoon Choi. Geoai: Integration of artificial intelligence, machine learning, and deep learning with gis. *Applied Sciences*, 13(6), 2023.
- [7] Rocio de la Torre, Canan G. Corlu, Javier Faulin, Bhakti S. Onggo, and Angel A. Juan. Simulation, optimization, and machine learning in sustainable transportation systems: Models and applications. *Sustainability*, 13(3), 2021.

- [8] Michael Dorosan, Damian Dailisan, Jesus Felix Valenzuela, and Christopher Monterola. Use of machine learning in understanding transport dynamics of land use and public transportation in a developing city. *Cities*, 144:104587, 2024.
- [9] Zohra Ayat Erahmane Fenghour, Djamel Raham, and Salheddine Sadouni. Dynamic spatial approach using gis and ai for enhanced accessibility of public transportation system: case of study ali mendjeli, algeria. *Euro-Mediterranean Journal for Environmental Integration*, 10(5):4265–4285, Oct 2025.
- [10] Laaziza Hammoumi, Saad Farah, Mohamed Benayad, Mehdi Maanan, and Hassan Rhinane. Leveraging machine learning to predict traffic jams: Case study of casablanca, morocco. *Journal of Urban Management*, 14(3):813–826, 2025.
- [11] Runpeng Liu and Seong-Yoon Shin. A review of traffic flow prediction methods in intelligent transportation system construction. *Applied Sciences*, 15(7), 2025.
- [12] Ali Louati. Machine learning framework for sustainable traffic management and safety in alkharj city. *Sustainable Futures*, 9:100407, 2025.
- [13] Hongrong Yang and Markus Schlaepfer. Urbanpulse: A cross-city deep learning framework for ultra-fine-grained population transfer prediction, 2025.